

California Housing Price Prediction

Model Interpretation Report

Project Proof of Concept

Dataset: featured_housing.csv | Rows: 20,640 | Features: 22

1. Dataset Overview

The dataset contains 20,640 California housing block-group records with 22 features including raw, engineered, and one-hot encoded variables. The prediction target is median_house_value.

Feature categories

- Geographic: longitude, latitude
- Housing stock: housing_median_age, total_rooms, total_bedrooms, households
- Population: population, population_per_household, rooms_per_household, bedrooms_per_room
- Income: median_income, log_median_income, income_category_Medium/High/Very High
- Log-transformed: log_total_rooms, log_population
- Ocean proximity (one-hot): INLAND, ISLAND, NEAR BAY, NEAR OCEAN

2. Model Performance Comparison

Three regression models were trained on an 80/20 train-test split (random_state=42). Results on the held-out test set:

Model	R ²	MAE (\$)	RMSE (\$)
Linear Regression	0.573	50,388	74,781
Random Forest	0.807	32,335	50,339
Gradient Boosting ★ Best	0.809	33,300	50,002

Key findings:

- Tree-based models (RF and GB) outperform Linear Regression substantially, reducing MAE by ~41% (\$50k vs \$32-33k).
- Gradient Boosting edges out Random Forest by a slim margin (R² 0.809 vs 0.807), making it the best-performing model.
- Linear Regression's lower R² (0.573) signals non-linear relationships in the data that a linear model cannot capture.

3. Feature Importance Analysis

3.1 Random Forest Feature Importances (top 10)

Rank	Feature	Importance (%)
1	median_income	25.5%

2	log_median_income	22.8%
3	ocean_proximity_INLAND	13.7%
4	population_per_household	12.1%
5	longitude	5.7%
6	latitude	5.6%
7	housing_median_age	4.4%
8	rooms_per_household	2.5%
9	bedrooms_per_room	2.4%
10	total_bedrooms	1.0%

3.2 Gradient Boosting Feature Importances (top 10)

Rank	Feature	Importance (%)
1	log_median_income	30.0%
2	median_income	23.0%
3	ocean_proximity_INLAND	14.8%
4	population_per_household	11.4%
5	longitude	6.9%
6	latitude	6.1%
7	housing_median_age	3.7%
8	rooms_per_household	1.3%
9	bedrooms_per_room	1.0%
10	total_bedrooms	0.4%

Interpretation:

- Income (median_income + log_median_income) together account for ~45-53% of feature importance across both tree models, making it the dominant predictor.
- ocean_proximity_INLAND is the strongest categorical feature (~14%), reflecting the well-known coastal premium in California real estate.
- population_per_household (~11-12%) acts as a crowding/density proxy and is the fourth most important feature.
- Geographic coordinates (lat/lon, ~11%) contribute but are partially subsumed by the ocean proximity features.

4. Linear Regression Coefficients

Coefficients are on standardised (z-scored) features, making their magnitudes directly comparable across features.

Rank	Feature	Coefficient (\$)	Direction
1	log_total_rooms	+63,489	Positive
2	latitude	-61,986	Negative
3	log_population	-61,668	Negative
4	longitude	-59,871	Negative
5	median_income	+36,311	Positive
6	log_median_income	+31,056	Positive
7	bedrooms_per_room	+27,784	Positive
8	households	+26,968	Positive
9	housing_median_age	+15,259	Positive
10	ocean_proximity_INLAND	-14,999	Negative

Caution: Several features are log-transformed versions of raw features (e.g., log_median_income and median_income). This multicollinearity inflates individual coefficient magnitudes and makes direct interpretation unreliable. The tree model importances are more trustworthy for causal interpretation.

5. Partial Dependence Analysis (Random Forest)

Partial Dependence Plots (PDPs) show the marginal effect of each feature on the predicted house value, averaging out all other features.

5.1 Median Income

- Values rise gradually from ~\$167k (income=1.6) to ~\$210k (income=5.0).
- A sharp non-linear jump occurs above income=5: values leap to ~\$305k at income=7.3.
- This step-change reveals a luxury premium in high-income neighbourhoods that a linear model cannot represent.

5.2 Housing Median Age

- Mild upward trend: new builds predict ~\$196k while 52-year-old stock predicts ~\$227k.
- Likely reflects that older properties sit in established, desirable neighbourhoods rather than structural age adding intrinsic value.

5.3 Population per Household

- Strong negative relationship: moving from 1.87 to 4.33 people/household drops predicted value from \$275k to \$172k (a \$103k decline).
- Overcrowded census blocks are a powerful negative signal, likely proxying for lower-income or high-density rental areas.

5.4 Longitude

- Far-west coastal areas (around -122.5°, Bay Area) command ~\$250k predictions.
- Values decline heading east toward -117° (Inland Empire/desert regions), reflecting the classic coast-to-inland price decay in California.

6. Summary and Recommendations

Area	Finding
Best model	Gradient Boosting ($R^2 = 0.809$, MAE = \$33,300)
Top predictor	Median income (raw + log) explains ~45-53% of variance
Location effect	INLAND flag and lon/lat together show clear coastal premium
Non-linearity	Income has a step-change luxury premium above income=5; crowding sharply depresses value
Linear model limits	41% higher MAE than tree models; multicollinearity distorts coefficients

For production deployment, Gradient Boosting is recommended. Future improvements could include SHAP values for individual-level explanations and interaction terms for the income x coastal proximity interaction.